Name: Yanchen Liu

USC ID: 4908758800

Read the following rules carefully:

- Write your name and ID number in the solution you submit.
- Please sign and submit the code of honor in the exam with your solutions. The exam cannot be graded without a signed code of honor. You are supposed to do all of the problems on your own without receiving help from others.
- Cheating in the exam will not be tolerated. If you are caught cheating, you will be reported to the authorities. The recommendation of the instructor will be at least an F in the course in such cases.
- Do not post any questions on Piazza about the exam. The TAs have been instructed to refrain from answering questions about the midterm. In case of ambiguity or problems in the questions, just do your best.
- The use of generative AI is prohibited in answering the questions.
- Problems are not sorted in terms of difficulty. Please avoid guess work and long and irrelevant answers.
- Instructions on submitting the solutions to paper and pencil and coding problems will be provided shortly by the TAs. You can handwrite or typeset your solutions to paper and pencil problems.
- Show all your work and your final answer. Showing only the final answer of a question may not receive full credit and you must show your solution and reasoning behind the answer. Simplify your answer as much as you can.
- The exam has 8 questions, 16 pages, and a total of 100 points.
- The submission deadline for this midterm is 11:59 PM, Saturday 24, 2025.
- As this is a take home exam that extends over several days, OSAS students DO NOT receive extra time, per OSAS guidelines.
- Any change in the midterm (paper and pencil or coding) after the deadline is considered late submission. One second late is late. The midterm is graded based on *when it was submitted, not when it was finished.* The midterm can be submitted up to two days late, with 10% penalty per late day. Homework late days cannot be used for the midterm.
- Submission after the grace period will receive a zero. One second late is late.

Grading Breakdown

Problem	Score	Earned
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	20	
8	20	
Total	100	

Honor Code

I pledge on my honor that I have not given or received any unauthorized assistance on this examination.

Name: Yanchen Liu

Signature: 

1. Although in practice, we usually model conditional distributions of features in each class as Gauusians for Bayesian Discriminant Analysis, one must notice that Bayesian Discriminant Analysis is doable using any type of distribution. Assume that we wish to perform Discriminant Analysis with only one positive predictor $x \geq 0$ (i.e. $p = 1$), but instead of normal class pdfs, we have Lognormal class pdfs whose formula is:

$$f_k(x) = \frac{1}{x\sqrt{2\pi}\sigma_k} \exp\left(\frac{-(\ln x - \mu_k)^2}{2\sigma_k^2}\right), x \geq 0$$

We know that the posterior class probability for class k is:

$$\Pr(Y = k|X = x) = p_k(x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

where π_l's are prior class probabilities.

(a) For $K = 3$ and $\pi_1 = \pi_2 = \pi_3$ and $\sigma_1 = 1, \sigma_2 = 2, \sigma_3 = 3$ and $\mu_1 = 1, \mu_2 = 5, \mu_3 = 7$, determine what ranges of x are classified in each class. Hint: plotting the pdfs may help.

(b) For binay classification ($K = 2$), when classes are balanced ($\pi_1 = \pi_2$), show that when $\sigma_1 = \sigma_2$, the decision boundary between the two classes is the threshold

$$x^* = \sqrt{\exp(\mu_1)}\sqrt{\exp(\mu_2)}$$

(a) Since the pirors are equal, classify x by maximizing $f_k(x)$. Let $t = \ln(x)$. Taking logarithms and removing terms common to all closes gives

$$g_1(t) = -\frac{(t-1)^2}{2}, \quad g_2(t) = -\ln 2 - \frac{(t-5)^2}{8}, \quad g_3(t) = -\ln 3 - \frac{(t-7)^2}{18}$$

$$\begin{cases} g_1 = g_2 \Rightarrow t \approx -3.326576, 2.659910 \\ g_1 = g_3 \Rightarrow t \approx -2.494882, 2.994882 \\ g_2 = g_3 \Rightarrow t \approx -0.005686, 6.805686 \end{cases} \Rightarrow \hat{Y}(x) = \begin{cases} 3, & 0 < x < 0.082506 \\ 1, & 0.082506 < x < 14.294998 \\ 2, & 14.294998 < x < 902.967049 \\ 3, & x > 902.967049 \end{cases}$$

(b) With equal pirors and $\sigma_1 = \sigma_2 = \sigma_3$, the deision boundary satisfies

$$f_1(x^*) = f_2(x^*) \Rightarrow (\ln x^* - \mu_1)^2 = (\ln x^* - \mu_2)^2$$

Thus: $(\mu_2 - \mu_1)(2\ln x^* - \mu_1 - \mu_2) = 0$

For $\mu_1 \neq \mu_2$: $x^* = e^{(\mu_1+\mu_2)/2}$

If $\mu_1 = \mu_2$, the distributions are identical and all points are ties.

2. Consider the following unusual logistic regression model for binary classification:

$$\Pr(Y=1|X_1,X_2)=\frac{\beta_0|X_1|^{\beta_1}|X_2|^{\beta_2}}{1+\beta_0|X_1|^{\beta_1}|X_2|^{\beta_2}}$$

we wish to fit the model to the following dataset:

Index	X_1	X_2	Y
1	1	-1	1
2	1	1	1
3	-1	1	0
4	-1	-1	0

(a) Write down the log odds model for this unusual logistic regression model.

(b) Write down the *likelihood function* for this dataset.

(c) Analytically find the parameters β_i that maximize the likelihood function (or minimize the negative log likelihood function.)

(a) Let $p=Pr(Y=1|X_1,X_2)$. Then:

$$\log\frac{p}{1-p}=\log\beta_0+\beta_1\log|X_1|+\beta_2\log|X_2|,\ \beta_0>0$$

(b) Since $|x_{i1}|=|x_{i2}|=1$, all observation have

$$p_i=\beta_0/(1+\beta_0)$$

Thus $L(\beta_0,\beta_1,\beta_2)=\prod_{i=1}^{4}p_i^{y_i}(1-p_i)^{1-y_i}=\beta_0^2/(1+\beta_0)^4$

(c) $l=2\log\beta_0-4\log(1+\beta_0)$,

$$\frac{\partial l}{\partial\beta_0}=\frac{2}{\beta_0}-\frac{4}{1+\beta_0}=\frac{2(1-\beta_0)}{\beta_0(1+\beta_0)}$$

The derivative is positive for $0<\beta_0<1$ and negative for $\beta_0>1$, Therefor:

$$\hat{\beta}_0=1,\ \hat{\beta}_1,\hat{\beta}_2\in\mathbb{R}$$

The likelihood does not depend on β_1,β_2, so they are unidentifiable.

The $L_{max}=1/16$.

3. The regularized least squares problem has the following objective function:

$$J(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda R(\beta_0, \beta_1, \ldots, \beta_p)$$
$$= \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} - \cdots - \beta_p x_{ip})^2 + \lambda R(\beta_0, \beta_1, \ldots, \beta_p)$$

Determine which of the following choices of $R(\beta_0, \beta_1, \ldots, \beta_p)$ are *valid* regularizers, in the sense of shrinking the parameters, and explain your answer. Also explain among the valid regularizers, which of them tends to yield sparse solutions for β_i's. Hint: for simplicity, you can think about each of them when applied to a regression problem with only one regressor.

(a) $R(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} \beta_i^3$

(b) $R(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} \ln(1 + |\beta_i|)$

(c) $R(\beta_0, \beta_1, \ldots, \beta_p) = \|\beta\|_0)$, where the $\mathscr{L}_0$ norm of the vector β is defined as the number of non-zero elements in $\beta = (\beta_0, \beta_1, \ldots, \beta_p)$. For example, if $\beta = (1, 2, 3, 4)$, then $\|\beta\|_0) = 4$ and for $\beta = (1000, 240, 0.03, 0)$, $\|\beta\|_0 = 3$, and for $\beta = (10^{-6}, 0.2, 0, 0)$, $\|\beta\|_0 = 2$.

(d) $R(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} \sin(\beta_i)$

(e) $R(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} |\beta_i|$

(f) $R(\beta_0, \beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} \beta_i^4$

(a) Invalid, β_i^3 becomes arbitrarily negative as $\beta_i \to -\infty$, making the objective unbounded below

(b) Valid and promotes sparsity: $\ln(1+|\beta_i|)$ increases with coefficient magnitude and has a kink at zero that encourages exact zeros

(c) Valid and promotes sparsity: $\|\beta\|_0$ directly penalizes nonzero coefficients, though retained coefficients need not shrink.

(d) Invalid: $\sin(\beta_i)$ oscillates and is not minimized at zero, so it does not consistently encourage shrinkage.

(e) Valid and promotes sparsity: $|\beta_i|$ penalizes large magnitudes and has a kink at zero that encourages exat zeros.

(f) Valid but generally does not promote sparsity: β_i^4 penalizes large magnitudes, but its smoothness and zero derivate at zero prevent exact zeros.

4. Consider the following multiclass dataset:

Index	X_1	X_2	Y
1	12	-1	+
2	-3	0	-
3	2	4	-
4	12	1	*
5	-5	17	-
6	11	8	+
7	13	10	+
8	-6	-1.5	*
9	21	-6	+

For this dataset, determine the leave-one-out cross validation estimate of the misclassification error of a simple classifier that *classifies to the prior*, i.e always classifies to the most common class in its training set. If you ever encounter a tie, break it in favor of * and then +.

The class counts are $n_+ = 4, n_- = 3, n_* = 2$

Leaving out a +: the counts become (3,3,2), so the tie-breaking rule selects +; all 4 such predictions are correct.

Leaving out a -: + remains the majority; all 3 such predictions are incorrect.

Leaving out a *: + remains the majority; both such predictions are incorrect.

Thus: LOOCV misclassification error $= \frac{3+2}{9} = 5/9$

5. Consider the task of identifying alligators and crocodiles in a pond based on their respective lengths. The average length of an adult crocodile is about 15 feet, while the average length of an adult alligator is about 12 feet. Assuming that their length X follows a Gaussian distribution with a standard deviation σ equal to 2 feet: (20 pts)

   (a) Write down the conditional pdf of X given being a crocodile.

   (b) Write down the conditional pdf of X given being an alligator.

   (c) Assuming that their prior probabilities are the same, the ideal decision boundary (threshold for X that distinguishes between two classes) is located at some length x^* such that $P(\text{Crocodile}|X = x^*) = P(\text{Alligator}|X = x^*)$.

   (d) Repeat 5c when we know that the population of the crocs is twice as much the population of alligators in the pond.

   (e) Repeat 5c when we know that the population of the alligators is twice as much the population of crocs in the pond.

   (f) Repeat 5c when the prior probabilities are the same, but standard deviations are different; i.e., $\sigma(\text{Crocodile}) = 4$ and $\sigma(\text{Alligator}) = 2$.

Let C denote crocodile and A denote alligator

(a) $f_C(x) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(x-15)^2}{8}\right)$

(b) $f_A(x) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(x-12)^2}{8}\right)$

(c) $f_C(x^*) = f_A(x^*) \Rightarrow (x^*-15)^2 = (x^*-12)^2 \Rightarrow x^* = 13.5$ ft

(d) $\pi_C = 2/3$, $\pi_A = 1/3$, $2f_C(x^*) = f_A(x^*) \Rightarrow \ln 2 + \frac{6x^*-81}{8} = 0$

$x^* = 13.5 - \frac{4}{3}\ln 2 \approx 12.576$ ft

(e) $\pi_C = 1/3$, $\pi_A = 2/3$, $\frac{6x^*-81}{8} - \ln 2 = 0$

$x^* = 13.5 + \frac{4}{3}\ln 2 \approx 14.424$ ft

For (c)–(e), predict crododile above the threshold and alligator below it

(f) With equal priors, $\sigma_C = 4$, $\sigma_A = 2$

$$\frac{1}{4\sqrt{2\pi}} e^{-(x-15)^2/32} = \frac{1}{2\sqrt{2\pi}} e^{-(x-12)^2/8}$$

$$\Rightarrow x = 11 \pm \sqrt{4 + \frac{32}{3}\ln 2} \approx 7.625, 14.375 \text{ ft}$$

6. The following least squares liner regression model was fitted to a sample of 28 students using data obtained at the end of their senior year in college. The aim was to explain students' weight gains:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3$$
$$\hat{y} = 6.15 + 0.553 x_1 - 1.141 x_2 + 0.813 x_3$$

The standard errors are:

$$\text{SE}(\hat{\beta}_1) = 0.165$$
$$\text{SE}(\hat{\beta}_2) = 0.365$$
$$\text{SE}(\hat{\beta}_3) = 0.202$$

- $\hat{y}$: weight gained, in pounds, during senior year
- x_1: average number of meals eaten per week
- x_2: average number of hours of exercise per week
- x_3: average number of beers consumed per week

(a) Interpret the estimated coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$

(b) Test, at the 1% level, the null hypothesis that the true coefficient on the variable x_3 is 0 against the alternative that it is not 0.

(c) Find and interpret a 94% confidence interval for the parameter β_1.

(d) If for the model, SSR=83.4 (Regression Sum of Squares) and SSE = 43.9 (Residual Sum of Squares), test the hypothesis that all the coefficients of the model are 0 (test overall significance of the model) using $\alpha = 3\%$. .

(a) $\hat{\beta}_0 = 6.15$: predicted senior-year weight gain is 6.15 pounds when all three predictors equal zero.
$\hat{\beta}_1 = 0.553$: each additional meal per week is associated with 0.553 pounds more senior-year weight gain, holding exercise and beer consumption fixed.

(b) $H_0: \beta_3 = 0$, $H_a: \beta_3 \neq 0$. $t = \frac{0.813}{0.202} = 4.025$, $df = 28-3-1 = 24$
Since $|t| = 4.025 > t_{0.995,24} = 2.797$, reject H_0 at the 1% level; beer consumption is a significant predictor.

(c) The 94% confidence interval is
$\hat{\beta}_1 \pm t_{0.97,24} SE(\hat{\beta}_1) = 0.553 \pm 1.974(0.165) = (0.227, 0.879)$
We are 94% confident that each additional meal per week is associated with an increase of 0.227 - 0.879 pounds in expected senior-year weight gain, holding the other predictor fixed.

(d) $H_0: \beta_1 = \beta_2 = \beta_3 = 0$,
H_0: at least one slope is non-zero

$$F = \frac{SSR/3}{SSE/24} = \frac{83.4/3}{43.9/24} = 15.198$$

Since $15.198 > F_{0.97;3,24} = 3.530$, reject H_0; the model is overall significant at the 3% level

7. The conditional probability distribution function of the weight W (in kg) given the height H (in cm) in a population is Gaussian (normal), and (35 pts)

$$p_{W|H}(w|h) = \frac{1}{\sqrt{2\pi} \times 10} \exp\left(-\frac{(w - 0.5 \times h^{1.001})^2}{200}\right)$$

(a) What is the best estimate of the weight of a person as a function of their height in the sense of *mean squared error*? (Hint: look up the Gaussian distribution, its mean, and variance. This problem is asking you to determine the regression function $w = f(h)$, which was shown in the lecture to be a statistical property of the conditional distribution of the output W given a particular value h of the input H.)

(b) Use the result you obtained in 7a to estimate the weight of people whose heights are 155, 165, and 190 cm.

(c) Would your answer to 7a and 7b change if the conditional variance of W given $H = h$ were a function of h, say, $\sigma(h)$, i.e.:

$$p_{W|H}(w|h) = \frac{1}{\sqrt{2\pi} \times \sigma(h)} \exp\left(-\frac{(w - 0.5 \times h^{1.001})^2}{2[\sigma(h)]^2}\right)$$

(d) Assume that instead of the conditional distribution, you have the following sample. Estimate the weight of people whose heights are 150, 155, 165, and 190 cm, using KNN with $k = 3$:

$$\hat{y}_{KNN} = \frac{y_1 + y_2 + \cdots + y_k}{k}$$

where $y_1, y_2, \cdots, y_k$ are the labels of the k nearest neighbors to your test instance.

Person	Height (cm)	Weight (kg)
1	171	80
2	168	78
3	191	100
4	182	80
5	150	65
6	178	83

(e) Repeat 7d, but instead of using the simple average of the labels of k nearest neighbors, which is use the following weighted average:

$$\hat{y}_{KNN} = \frac{w_1 y_1 + w_2 y_2 + \cdots + w_k y_k}{w_1 + w_2 + \cdots + w_k}$$

where the weight w_i for the label y_i of instance i is determined as $1/d_i$, where d_i the distance between the instance i and the test instance. **It is worth noting**

that normalized weights can be viewed as similarities of the training instances with the test instance. An alternative formula would be:

$$\hat{y}_{KNN} = s_1 y_1 + s_2 y_2 + \cdots + s_k y_k$$

where $s_i = sim(x_i, x^*)$, where x^* is the test point.

(a) Since $W|H = h \sim N(0.5h^{1.001}, 100)$ the predictor minimizing mean squared error is the conditional mean: $\hat{w}(h) = E[W|H=h] = 0.5h^{1.001}$

(b) $\hat{w}(155) = 77.89$, $\hat{w}(165) = 82.92$, $\hat{w}(190) = 95.50$ kg

(c) No. The conditional mean remains $0.5h^{1.001}$, so the optional predictor and all three estimates are unchanged.

(d) Using the three nearest heights:

Height (cm)	Nearest Height (cm)	Estimated Weight (kg)
150	150, 168, 171	(65+78+80)/3 = 74.33
155	150, 168, 171	(65+78+80)/3 = 74.33
165	168, 171, 178	(78+80+83)/3 = 80.33
190	191, 182, 178	(100+80+83)/3 = 87.67

(e) Using the same neighbors with inverse-distance weights:

$$\hat{w}(155) = \frac{65/5 + 78/13 + 80/16}{1/5 + 1/13 + 1/16} = 70.71 \text{ kg}$$

$$\hat{w}(165) = \frac{78/3 + 80/6 + 83/13}{1/3 + 1/6 + 1/13} = 79.24 \text{ kg}$$

$$\hat{w}(190) = \frac{100/1 + 80/8 + 83/12}{1 + 1/8 + 1/12} = 96.76 \text{ kg}$$

at $h = 150$, one distance is zero, making $1/d$ undefined. Using the standard exact-match rule (equivalently, the limiting prediction):

$$\hat{w}(150) = 65 \text{ kg}$$

8. **Programming Question: Predicting Housing Prices with Linear Regression**

   Dataset: `student-mat.csv`

   Link: https://archive.ics.uci.edu/dataset/320/student+performance

   In this problem, you will build a linear regression model to predict housing prices in the USA based on various features. You will use Python and `scikit-learn` for this task.

   (a) **Data Exploration and Pre-processing**

   i. Import the `student-mat.csv` dataset as a pandas DataFrame.

   ii. Select the following features: `age, studytime, schoolsup, goout, Dalc, Walc, health, absences, G3` (target/dependent variable). Encode the binary variable (`schoolsup`) values as 0s (no) and 1s (yes). Combine `Dalc` and `Walc` into one variable `alc` by taking averge, then remove `Dalc` and `Walc`. Display the first five rows of the pre-processed dataset.

   iii. Find the number of outliers for each independent variable using the IQR method.

   iv. Standardize and run PCA on the dataset. Create a scatterplot of PC1 vs PC2, coloring the dots by their final grade `G3`. Inspect the component loadings and determine which features contribute the most to PC1 and PC2. *Keep and use standardized data for remaining problems.*

   (b) **Linear Regression**

   i. Split data into training set and testing set with an 80:20 ratio. Use random seed 552 for reproducibility.

   ii. Build three models using the training set: A. Linear Regression Model, B. Linear Regression Model with Ridge Regularization, and C. Linear Regression Model with Lasso Regularization. Set α=0.1.

   iii. Test all three models on the test set. Find out the best performing model with repsect to each of the metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2.

   iv. How do you interpret R^2 values from the three models?

   v. Print coefficients of independent variables from the three models in one table.

   vi. What's the relationship between each independent variable and the dependent variable?

   vii. How do the regularization methods differ? What can you conclude about the dataset and features given the results?

   viii. If the regularization strength (α) is increased, what would happen to performance metrics?

   ix. What are some feature engineering methods you would suggest to improve model performance?

   Expected Output: Your submission should include:

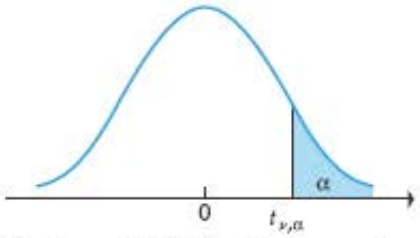
- Jupyter Notebook `.ipynb` with all the steps clearly commented.
- The output of each step as specified above (e.g., head of DataFrame, info, describe, missing value counts, evaluation metrics for all models, coefficients, and intercept).
- Visualizations for outlier detection and residual analysis.
- A brief discussion answering the interpretation questions.

Cumulative Distribution Function, $F(z)$, of the Standard Normal Distribution Table

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09

Cumulative Distribution Function, $F(z)$, of the Standard Normal Distribution Table

Upper Critical Values of Student's t Distribution with ν Degrees of Freedom



For selected probabilities, α, the table shows the values $t_{\nu,\alpha}$ such that $P(t_\nu > t_{\nu,\alpha}) = \alpha$, where t_ν is a Student's t random variable with ν degress of freedom. For example, the probability is .10 that a Student's t random variable with 10 degrees of freedom exceeds 1.372.

	Probability of Exceeding the Critical Value					
ν	0.10	0.05	0.025	0.01	0.005	0.001
1	3.078	6.314	12.706	31.821	63.657	318.313
2	1.886	2.920	4.303	6.965	9.925	22.327
3	1.638	2.353	3.182	4.541	5.841	10.215
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.893
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.782
8	1.397	1.860	2.306	2.896	3.355	4.499
9	1.383	1.833	2.262	2.821	3.250	4.296
10	1.372	1.812	2.228	2.764	3.169	4.143
11	1.363	1.796	2.201	2.718	3.106	4.024
12	1.356	1.782	2.179	2.681	3.055	3.929
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
16	1.337	1.746	2.120	2.583	2.921	3.686
17	1.333	1.740	2.110	2.567	2.898	3.646
18	1.330	1.734	2.101	2.552	2.878	3.610
19	1.328	1.729	2.093	2.539	2.861	3.579
20	1.325	1.725	2.086	2.528	2.845	3.552
21	1.323	1.721	2.080	2.518	2.831	3.527
22	1.321	1.717	2.074	2.508	2.819	3.505
23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450
26	1.315	1.706	2.056	2.479	2.779	3.435
27	1.314	1.703	2.052	2.473	2.771	3.421
28	1.313	1.701	2.048	2.467	2.763	3.408
29	1.311	1.699	2.045	2.462	2.756	3.396
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
60	1.296	1.671	2.000	2.390	2.660	3.232
100	1.290	1.660	1.984	2.364	2.626	3.174
∞	1.282	1.645	1.960	2.326	2.576	3.090
ν	0.10	0.05	0.025	0.01	0.005	0.001

F - Distribution ($\alpha = 0.01$ in the Right Tail)

Denominator Degrees of Freedom df_2 \ Numerator Degrees of Freedom df_1	1	2	3	4	5	6	7	8	9
1	4052.2	4999.5	5403.4	5624.6	5763.6	5859.0	5928.4	5981.1	6022.5
2	98.503	99.000	99.166	99.249	99.299	99.333	99.356	99.374	99.388
3	34.116	30.817	29.457	28.710	28.237	27.911	27.672	27.489	27.345
4	21.198	18.000	16.694	15.977	15.522	15.207	14.976	14.799	14.659
5	16.258	13.274	12.060	11.392	10.967	10.672	10.456	10.289	10.158
6	13.745	10.925	9.7795	9.1483	8.7459	8.4661	8.2600	8.1017	7.9761
7	12.246	9.5466	8.4513	7.8466	7.4604	7.1914	6.9928	6.8400	6.7188
8	11.259	8.6491	7.5910	7.0061	6.6318	6.3707	6.1776	6.0289	5.9106
9	10.561	8.0215	6.9919	6.4221	6.0569	5.8018	5.6129	5.4671	5.3511
10	10.044	7.5594	6.5523	5.9943	5.6363	5.3858	5.2001	5.0567	4.9424
11	9.6460	7.2057	6.2167	5.6683	5.3160	5.0692	4.8861	4.7445	4.6315
12	9.3302	6.9266	5.9525	5.4120	5.0643	4.8206	4.6395	4.4994	4.3875
13	9.0738	6.7010	5.7394	5.2053	4.8616	4.6204	4.4410	4.3021	4.1911
14	8.8616	6.5149	5.5639	5.0354	4.6950	4.4558	4.2779	4.1399	4.0297
15	8.6831	6.3589	5.4170	4.8932	4.5556	4.3183	4.1415	4.0045	3.8948
16	8.5310	6.2262	5.2922	4.7726	4.4374	4.2016	4.0259	3.8896	3.7804
17	8.3997	6.1121	5.1850	4.6690	4.3359	4.1015	3.9267	3.7910	3.6822
18	8.2854	6.0129	5.0919	4.5790	4.2479	4.0146	3.8406	3.7054	3.5971
19	8.1849	5.9259	5.0103	4.5003	4.1708	3.9386	3.7653	3.6305	3.5225
20	8.0960	5.8489	4.9382	4.4307	4.1027	3.8714	3.6987	3.5644	3.4567
21	8.0166	5.7804	4.8740	4.3688	4.0421	3.8117	3.6396	3.5056	3.3981
22	7.9454	5.7190	4.8166	4.3134	3.9880	3.7583	3.5867	3.4530	3.3458
23	7.8811	5.6637	4.7649	4.2636	3.9392	3.7102	3.5390	3.4057	3.2986
24	7.8229	5.6136	4.7181	4.2184	3.8951	3.6667	3.4959	3.3629	3.2560
25	7.7698	5.5680	4.6755	4.1774	3.8550	3.6272	3.4568	3.3239	3.2172
26	7.7213	5.5263	4.6366	4.1400	3.8183	3.5911	3.4210	3.2884	3.1818
27	7.6767	5.4881	4.6009	4.1056	3.7848	3.5580	3.3882	3.2558	3.1494
28	7.6356	5.4529	4.5681	4.0740	3.7539	3.5276	3.3581	3.2259	3.1195
29	7.5977	5.4204	4.5378	4.0449	3.7254	3.4995	3.3303	3.1982	3.0920
30	7.5625	5.3903	4.5097	4.0179	3.6990	3.4735	3.3045	3.1726	3.0665
40	7.3141	5.1785	4.3126	3.8283	3.5138	3.2910	3.1238	2.9930	2.8876
60	7.0771	4.9774	4.1259	3.6490	3.3389	3.1187	2.9530	2.8233	2.7185
120	6.8509	4.7865	3.9491	3.4795	3.1735	2.9559	2.7918	2.6629	2.5586
∞	6.6349	4.6052	3.7816	3.3192	3.0173	2.8020	2.6393	2.5113	2.4073